Prediction of Anti-Cancer Drug Efficacy Scores

Single-Cell Drug-Response Prediction with Foundation-Model Embeddings

Preliminary Report

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Abstract

This report investigates whether single-cell RNA-seq, combined with foundation-model embeddings, can predict pharmacological response, as a first step toward a reusable pan-cancer single-cell foundation model. On the highest-confidence intersection of the SCP542 single-cell atlas and the CTRPv2 drug-response database (180 cell lines, 545 compounds), we train a shared-trunk multi-task regressor on individual cells and evaluate it under a leave-cell-line-out protocol, comparing scGPT embeddings against a matched principal component analysis (PCA) baseline. The guiding hypothesis is that the foundation model provides a denoised, tissue-agnostic representation of cellular state. With a per-drug-standardized area-under-the-curve target, which is required for the 545-head objective to be learnable, the model ranks unseen cell lines at a Spearman correlation of 0.328 across all drugs and up to 0.396 on a learnable subset, and it exceeds the field-standard NaiveMeanEffects baseline under the DrEval benchmark (normalized Spearman 0.357 for scGPT). Model capacity is not the limiting factor: a ridge regression on cell-line mean embeddings ($\rho = 0.343$) matches the deep model, which indicates that the ceiling is the small number of independent cell lines and the noise of the broadcast bulk labels rather than the representation or the architecture. The scGPT embeddings provide a small advantage over PCA that largely fades under DrEval's normalized metric. We conclude that the approach is viable but currently label-limited, and we outline label- and data-side directions for the next phase.

1 Introduction

The overarching goal of this project is to develop a domain-specific, pan-cancer single-cell foundation model capable of predicting pharmacological response, both efficacy and toxicity, that can subsequently be fine-tuned for specific cancer types or for clinical datasets with binary outcomes. Tumor heterogeneity is a major driver of drug resistance and treatment failure, yet this resolution is obscured whenever drug response is modeled from bulk RNA-seq. Predicting response at the single-cell level would allow a model to report a distribution of sensitivities within a sample and, in principle, to identify rare, naturally resistant subclones before selection pressure acts on them. The immediate obstacle is that single-cell drug-screening data does not exist at scale.

To make progress despite this, the project adopts a weakly supervised strategy in which bulk pharmacological labels are mapped onto the individual cells of the corresponding *in vitro* cell lines, and the model is trained to map heterogeneous single-cell inputs to the average bulk response. Laboratory cell lines do not fully recapitulate clinical tumors, but they provide the high-volume, continuous labels required to train a robust representation that can later be fine-tuned on scarcer clinical data. The central methodological difficulty in this setup is a resolution mismatch: standard pharmacological labels, including those of CTRPv2 [1], PRISM [2] and GDSC [3], are derived from bulk cell populations, whereas the transcriptomic data from SCP542 [4] captures individual cellular heterogeneity. Mapping one average bulk score onto hundreds of heterogeneous cells in-

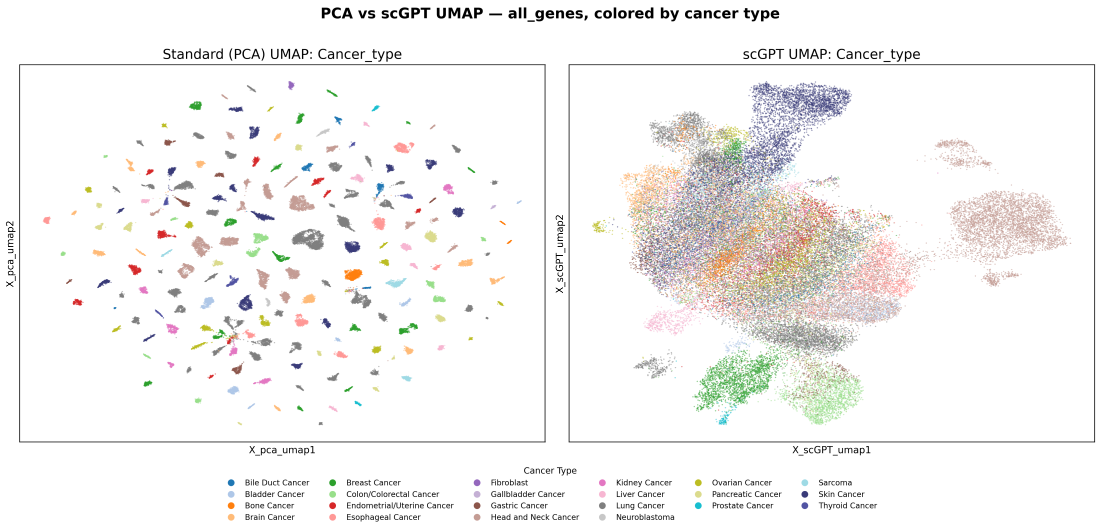

Figure 1: **scGPT removes the tissue-of-origin bias that dominates a PCA baseline.** Cells are embedded by full-transcriptome PCA (left) and by scGPT (right), reduced to two dimensions by UMAP and colored by cancer type. PCA separates the data into tissue-specific clusters, whereas scGPT projects the cells onto a continuous, shared manifold. Produced by `notebooks/06_verify_variants.ipynb`.

troduces substantial label noise, and regressing raw, sparse single-cell counts directly onto these labels is unstable.

This motivates the use of a foundation-model representation as an intermediate prior. An exploratory latent-space analysis illustrates the effect (Figure 1). When cells are colored by cancer type, full-transcriptome PCA is dominated by tissue-specific markers and separates the data into distinct lineage clusters, so that a downstream model could achieve low error simply by classifying the cell line. The scGPT embeddings [6] instead project the same cells onto a continuous, shared manifold in which tissues are intermixed. The project therefore diverges from prior work such as PERCEPTION [5], which pre-trained on bulk data to predict PRISM response, by focusing on reusable representation learning: it evaluates whether transformer-based single-cell embeddings improve drug-response prediction over a standard PCA baseline, and it is designed to extend to multi-task learning across several databases. The central hypothesis is that scGPT embeddings act as a denoised biological prior that aligns functional cellular states across tissues, so that a regressor trained on this representation is encouraged to learn transcriptomic drivers of sensitivity rather than to memorize the tissue of origin.

This report covers only the initial phase of the work: a single database (CTRPv2), a single continuous response score, single-cell inputs and a supervised regressor. The multi-task run reported here spans 545 drugs but a single database and metric; cross-database integration, toxicity and clinical fine-tuning are left to later work.

2 Methods

2.1 Data

All experiments use the highest-confidence intersection of SCP542 [4] and CTRPv2 [1]. This intersection comprises 180 cell lines with post-quality-control measurements (190 roster name matches, of which 10 are unscreened), 545 compounds and 53,513 cells, with effectively complete viability coverage inside the overlap. Genes are reduced to the 5,000 most highly variable; scGPT embeds the 4,576 of these that lie in its vocabulary, while PCA is computed on all 5,000. The gene-set size is not critical: an earlier sweep from 1,000 genes to the full transcriptome showed no advantage for any particular count (`notebooks/07`), so the 5,000-gene selection is used throughout.

A property of the data governs every error estimate in this report. The label is defined per cell line and drug, not per cell, so all cells of a line share the same broadcast bulk value and act as pseudo-replicates. The effective sample size is therefore the number of cell lines, approximately 150 after the held-out splits are removed, not the 53,513 cells. Treating cells as independent would understate the uncertainty by roughly the square root of the number of cells per line.

2.2 Representation and model

The comparison between representations is controlled so that only the content of the embedding differs. Both the PCA baseline (`X_pca`) and the scGPT embedding (`X_scGPT`) are 512-dimensional and are fed into the same multi-head perceptron, with hidden dimensions 128 and 64, layer normalization, GELU activations and dropout. The network has one output head per drug and is trained with a masked mean-squared-error loss that is evaluated only on observed cell-drug pairs. The model and training code are defined in `scripts/model/OncoMLP.py` and `scripts/training/train_multitask.py`.

2.3 Response target and per-drug standardization

The choice of training target requires care. CTRPv2 provides a dose-averaged percent-viability score, denoted `mean_pv`, and an integrated area under the fitted dose-response curve, `auc`. The initial experiments used `mean_pv`; the final target is the curve-fit AUC standardized per drug across cell lines, `auc_z`, defined for cell line i and drug j as $\text{auc\_z}_{ij} = (\text{auc}_{ij} - \mu_j)/\sigma_j$, where μ_j and σ_j are the mean and standard deviation of that drug's AUC over the cell lines.

Two considerations motivate the standardization. First, the per-drug response variance is highly heterogeneous across the 545 compounds, with the per-drug standard deviation ranging from 0.034 to 0.302, a factor of roughly 80 in squared error. In a shared multi-task objective with one masked squared-error term per drug, the contribution of a drug scales with its variance, so a small number of high-variance compounds dominate the shared representation irrespective of whether that variance carries any learnable signal; the three drugs with the largest spread (`ifosfamide`, `fqi-2` and `bafilomycin a1`) in fact fail to separate the cell lines at all. Standardizing each drug to unit variance is algebraically equivalent to weighting each head by $1/\sigma_j^2$, which removes this imbalance and is the regression analogue of loss reweighting under class imbalance [8]. Second, per-drug standardization changes the question from absolute potency, which is a property of the drug, to the relative sensitivity of a cell line for a given drug, which is the quantity of interest for personalized prediction. Because the Spearman correlation within a drug is invariant to an affine per-drug transform, the standardization does not alter any single-drug evaluation metric; its only effect is on the multi-task coupling. The target is implemented in `scripts/preprocessing/ctrp_to_h5ad.py`.

2.4 Evaluation

Cell lines are evaluated out of fold. A 5-fold group k-fold split, grouped by cell line, is applied to the 153 training and validation lines, so that no cell of a test line is seen during training. Each line is predicted exactly once, the per-cell predictions of a line are averaged to the line level, and the Spearman correlation between predicted and true values across the roughly 150 lines is computed per drug and averaged over drugs. This protocol replaces an earlier fixed 27-line validation split, on which the standard error of a correlation is approximately 0.2 and effects below 0.4 are not measurable. The Pearson correlation tracks the Spearman value within 0.02 throughout.

For external comparison, the model is additionally evaluated with the DrEval package [7] (version

Table 1: Out-of-fold Spearman correlation, averaged over the 10 learnable drugs, seed 42. The three targets tie at K = 10, but only the standardized `auc_z` survives at K = 545. Values are means of the per-drug rows in `outputs/target/target_comparison.csv` (`notebooks/11`).

Heads	Representation	`mean_pv`	`auc`	`auc_z`
10	PCA	0.370	0.360	0.360
10	scGPT	0.396	0.405	0.396
545	PCA	+0.073	−0.012	0.316
545	scGPT	−0.078	−0.069	0.328

1.5.1), using its leave-cell-line-out splits, its baselines and its evaluation routine without reimplementation. As recommended by the framework, metrics are reported after the mean drug and cell-line effects have been removed from both the true and predicted responses, and they are pooled over all cell-line-drug pairs rather than averaged per drug, so they are not directly comparable to the per-drug correlations above.

3 Results

3.1 Per-drug standardization is what makes the multi-task objective learnable

Table 1 compares the three targets across representations and head counts, with all other settings fixed. On the subset of 10 learnable drugs the three targets are statistically indistinguishable, which shows that the curve fit itself buys no accuracy over the dose average. The difference appears only at the full 545-head scale: both unstandardized targets collapse to approximately zero, with scGPT turning negative, whereas the standardized target retains $\rho = 0.328$ (scGPT). Per-drug standardization is therefore a prerequisite for a learnable large multi-task objective rather than a change to the information content of the label.

A set of single-parameter interventions on the unstandardized 545-head configuration locates the cause (Figure 2). Starting from the broken baseline ($\rho = -0.08$), heavier regularization, a smaller model, a different batch size and sample reweighting all leave the result unchanged or make it worse. Two changes help. Removing the regularization recovers most of the gap (to $\rho = +0.21$), which shows that the dropout had been interacting with the mis-scaled loss by starving the learnable heads of shared capacity. Standardizing the target recovers the signal fully (to $\rho = +0.33$) without weakening the regularization, which is why, once the target is corrected, regularization is flat and no longer a useful lever.

3.2 The corrected model ranks unseen cell lines at full head count

With the standardized target the model ranks unseen cell lines well above chance. On the learnable subset it reaches $\rho = 0.396$ (scGPT) and $\rho = 0.360$ (PCA), and it holds at $\rho = 0.328$ and $\rho = 0.316$ even when all 545 heads are trained jointly, which shows that the large multi-task objective is trainable once the target is standardized.

3.3 Model capacity is exhausted; the ceiling is the number of independent cell lines

On the corrected setting, model-side tuning yields nothing further: regularization, capacity (from 74,954 down to 5,130 parameters) and batch size all leave the correlation flat. The decisive control is a ridge regression fitted to the 150 cell-line mean embeddings, with no single cells and no network, which reaches $\rho = 0.343$ (PCA) and essentially ties the PCA multi-layer perceptron

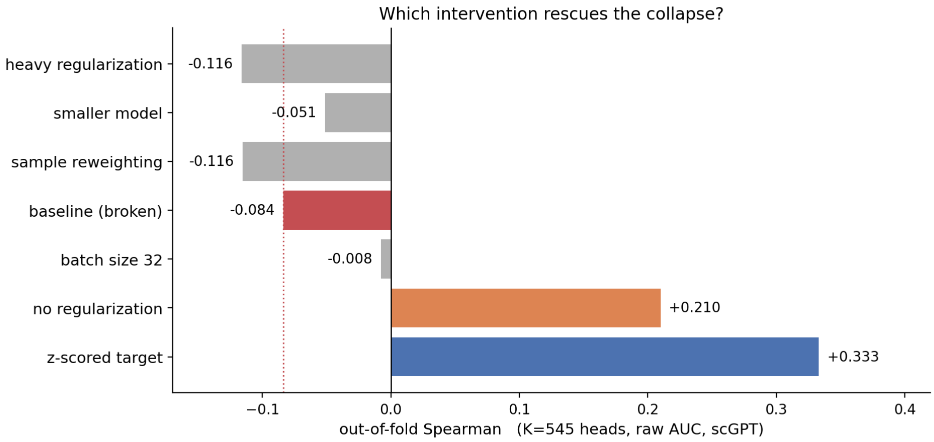

Figure 2: **On the broken 545-head configuration, only removing regularization or standardizing the target recovers the signal.** Each bar applies a single change to the same unstandardized setting (raw `auc`, scGPT). Heavier regularization, a smaller model, a different batch size and reweighting (grey) do not help; removing regularization (orange) recovers most of the gap; per-drug standardization (blue) recovers it fully and addresses the cause. Source: `outputs/ablations/rescue_k545.png` (`notebooks/10`).

Table 2: DrEval leave-cell-line-out benchmark, normalized metrics, mean over five folds. The model clears the NaiveMeanEffects baseline and edges DrEval's reference models on identical features. Data source: `outputs/dreval/dreval_lco_results.csv` (`notebooks/12`).

Model	norm. Spearman	norm. R^2
NaiveMeanEffects (baseline)	0.000	0.000
SingleDrugRandomForest (scGPT)	0.339	0.098
OncoMLP (X_pca)	0.340	0.086
OncoMLP (X_scGPT)	0.357	0.114

($\rho = 0.356$). The deep single-cell apparatus adds a measurable margin only through scGPT ($\rho = 0.402$), and even there the scGPT signal is nonlinear, since a linear head on scGPT drops to $\rho = 0.292$. The scGPT representation does, however, generalize better than PCA in the way the guiding hypothesis predicts: sharing an identical trunk, its training-to-validation gap on the corrected target is 0.098, against 0.464 for PCA, so the foundation-model embedding overfits far less. The limiting factor is thus the small number of independent cell lines and the noise of the broadcast labels, not the representation or the architecture.

3.4 Against the field, the model clears the naive baseline but not the best-in-class model

Under the DrEval leave-cell-line-out protocol and its normalized metric (Table 2, Figure 3), our model exceeds the NaiveMeanEffects baseline that half of the published field fails to beat, reaching a normalized Spearman correlation of 0.357 for scGPT against 0.000 for the baseline, and it slightly exceeds DrEval's own SingleDrug reference models on identical features ($\rho = 0.339$ for their random forest). It remains below the best model the DrEval study reports, whose normalized coefficient of determination reaches about 19% against our 0.114. The scGPT-over-PCA advantage seen in the internal evaluation shrinks to near noise once the mean effects are removed (normalized Spearman 0.357 versus 0.340). Consistent with the ridge control above, this reproduces the study's central finding that most of the apparent performance in this field is the cell-line and drug mean effect rather than differential response.

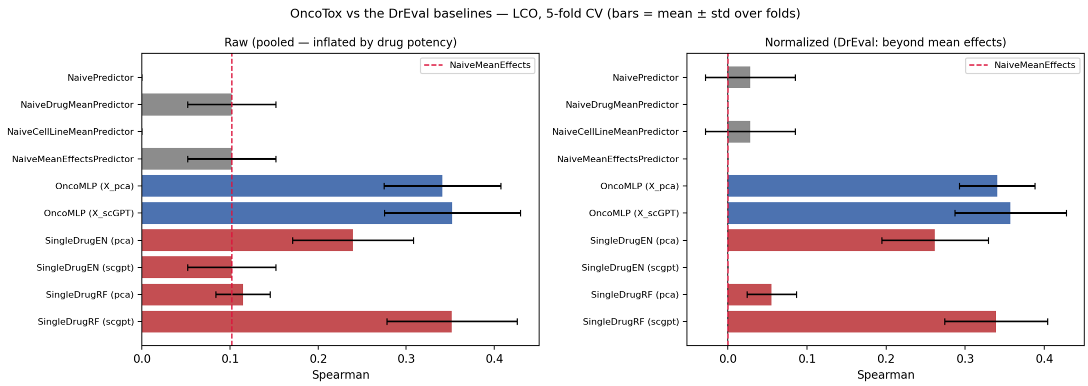

Figure 3: **The naive predictors' raw performance is bias that the normalization removes.** DrEval leave-cell-line-out results, raw pooled correlation (left) and normalized (right). The naive predictors sit above zero on the left and collapse to zero on the right, which shows that their raw performance was the mean effect; the model remains above the baseline after normalization. Source: `outputs/dreval/dreval_lco.png` (`notebooks/12`).

4 Discussion

The results reframe an earlier, apparently negative finding. A first pass of this work had reported that neither representation could rank cell lines, with out-of-fold correlations near zero over the full drug panel. That outcome was substantially an artifact of the training target rather than a statement about the data or the representation. The initial score, a dose-averaged viability, left the per-drug variances untouched, so in the shared multi-task loss a handful of high-variance and largely uninformative compounds dominated the representation. Standardizing the target per drug removes this imbalance, and with it the model ranks unseen cell lines at correlations of 0.328 across the panel and up to 0.396 on a learnable subset. The standardization is best understood as a necessary preprocessing choice for a heterogeneous multi-task objective, not as a scientific result in itself; a head-to-head comparison of the candidate scores is informative mainly because it shows that the curve fit alone changes nothing and the standardization changes everything. The dominance of the target change is direct: on the unstandardized targets the same model collapses at full head count (Table 1), and no other single intervention on the broken configuration approaches it (Figure 2).

The corrected picture is modest but genuine, and it is best read against the standards the DrEval study sets for the field. The model clears the NaiveMeanEffects baseline that about half of published models fail to beat, and it edges DrEval's own reference models on identical features, but it remains below the best model that study reports and shows only a faint representation advantage once the mean effects are normalized away. The most informative internal result points the same way: a ridge regression on cell-line mean embeddings matches the deep model, so the information the current pipeline exploits is largely captured by a per-line average. This is the cell-line-effect bias that DrEval was designed to expose, reproduced here independently. The single-cell resolution and the scGPT embedding together add a small margin over that average, and scGPT needs a nonlinear head to realize it, which is the first concrete evidence that the embedding carries structure a linear model cannot reach. This lower reliance on memorization is the effect the guiding hypothesis anticipated, and it persists on the corrected target, where the scGPT model's training-to-validation gap (0.098) is far smaller than that of PCA (0.464) under an identical trunk.

Taken together, the evidence indicates that the limiting factor at this stage is the label side rather than the model side. The task is defined on roughly 150 independent cell lines, each carrying one bulk value broadcast onto hundreds of cells, and model-side tuning is exhausted across regularization, capacity and batch size. Improving performance therefore requires acting on the labels and the data, not on the architecture.

5 Limitations and Future Work

Several limitations bound the strength of the present claims, and each points to a concrete next step.

The most serious is that the learnable-drug subset was selected using all 180 cell lines, including the validation and test lines, so any number reported on that subset is a best-case diagnostic rather than a generalization estimate; the selection should instead be performed inside each cross-validation fold, using only the training lines. The subset is moreover chosen from label statistics alone and was not validated in this report against the per-drug performance the model actually achieves, so it should be read as a diagnostic device rather than a claim that only ten drugs are learnable.

The scGPT advantage over PCA is small: a Spearman gap of +0.036 at $K = 10$ and +0.012 at $K = 545$, of the same order as the seed-to-seed variation. Across three seeds the gap averages +0.04 and is not sign-consistent, ranging from −0.00 to +0.09, and it does not survive DrEval's normalized metric, so it should not be presented as an established margin until it has been reproduced over more seeds and a wider drug set. The external benchmark is itself preliminary, covering only the leave-cell-line-out split on the learnable subset rather than the leave-tissue-out and leave-drug-out settings or the full panel.

The clearest opportunities for improvement follow from the diagnosis that the task is label-limited. First, the training target can be aligned with the evaluation by regressing on the doubly normalized residual, in which both the drug and the cell-line mean effects are removed, with the means estimated only on the training lines of each fold; this matches the quantity DrEval rewards, and an initial check (`outputs/dreval/dreval_normalized.csv`) indicates that about a fifth of the current signal is the cell-line effect alone. Second, the single-cell resolution has yet to justify itself, since averaging a line's cells loses nothing measurable; a pooling model that predicts the line label from a bag of cells, for example through attention, is the natural test and must beat the ridge control to be worthwhile. Third, and most directly, the ceiling can be raised by adding independent cell lines, as CTRPv2 alone screens roughly 1,100 against the 180 in the current overlap. Beyond these, a diagnostic error analysis of where the model fails is low-risk and immediately useful, whereas gene-level interpretation to surface transcriptomic drivers of resistance is worthwhile only once per-drug performance is substantially higher and stable, so that the interpretation does not merely describe noise. The longer horizon remains a cross-database, multi-task model spanning efficacy and toxicity, fine-tunable on clinical outcomes.

References

[1] Seashore-Ludlow, B. et al. Harnessing connectivity in a large-scale small-molecule sensitivity dataset. *Cancer Discovery* **5**, 1210–1223 (2015); and Rees, M. G. et al. Correlating chemical sensitivity and basal gene expression reveals mechanism of action. *Nature Chemical Biology* **12**, 109–116 (2016).

[2] DepMap, Broad; Kocak, M. Repurposing Public 24Q2. figshare dataset (2024). https://doi.org/10.6084/m9.figshare.25917643.v1

[3] Genomics of Drug Sensitivity in Cancer (GDSC), Wellcome Sanger Institute. https://cellmodelpassports.sanger.ac.uk/downloads

[4] Kinker, G. S. et al. Pan-cancer single-cell RNA-seq identifies recurring programs of cellular heterogeneity. *Nature Genetics* **52**, 1208–1218 (2020). https://doi.org/10.1038/s41588-020-00726-6

[5] Sinha, S. et al. PERCEPTION predicts patient response and resistance to treatment using single-cell transcriptomics of their tumors. *Nature Cancer* **5**, 938–952 (2024). https://doi.org/10.1038/s43018-024-00756-7

[6] Cui, H. et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods* **21**, 1470–1480 (2024). https://doi.org/10.1038/s41592-024-02201-0

[7] Bernett, J., Iversen, P., Picciani, M., Wilhelm, M., Baum, K. & List, M. Critical evaluation of drug response prediction models with DrEval. *Nature Communications* (2026). Software: `drevalpy` v1.5.1.

[8] Kendall, A., Gal, Y. & Cipolla, R. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2018).